

Non-Normal Performance Distributions in Rating Systems

with an application to seventy years of Formula 1

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Abstract

Rating systems almost universally assume that contest performance is ability plus normal or logistic noise. We present a setting where the assumption is visibly false and measurably costly: Formula 1, where for decades the most likely outcome of starting a race was not finishing it. A lattice-based Thurstonian rater that accepts arbitrary noise densities outperforms TrueSkill, Elo, Glicko-2 and OpenSkill over 1,158 races, and replacing its normal noise with a two-component density, a Gaussian pace term plus a separated block of slow mass representing retirement, improves every metric. Estimating the block’s mass from the trailing retirement rate improves it further. Symmetric heavy tails and skew do not help, so the gain is attributable to the block, not to non-normality in general.

1 Introduction

Nearly every rating system in use rests on the same distributional convention. A competitor’s performance in a contest is modeled as a skill parameter plus noise, and the noise is normal (Thurstone, TrueSkill, the Thurstone–Mosteller systems) or logistic or Gumbel (Elo, Glicko, Bradley–Terry, Plackett–Luce). The convention is so uniform that it is rarely stated as an assumption. This paper is about what happens when it is false, and about how much can be gained by replacing it, in a setting where the truth is not subtle.

That setting is Formula 1. The fraction of entrants failing to finish, by decade:

decade	1950s	60s	70s	80s	90s	2000s	10s	20s
DNF rate	.42	.49	.51	.59	.51	.30	.18	.13

For most of the sport’s history, retirement was the most likely outcome of starting a race. A retirement is not a slow day drawn from the tail of a pace distribution; it is a mechanical event whose result is unrelated to how fast the car was. Performance in this sport is self-evidently a mixture: pace when the car survives, and something categorically different when it does not. No normal, logistic or Gumbel density represents this, which makes Formula 1 a natural experiment for the value of freeing the noise distribution. The results below quantify that value and, just as usefully, show which departures from normality do not help.

Working with a free noise density requires a rating system that can accommodate one. Section 3 describes such a system and verifies it is a strong baseline in its own right. Section 4 contains the central experiment: the noise density sweep. Section 5 compares the resulting ratings with betting market prices, a secondary exercise limited by the surviving odds record.

2 Data and protocol

Race results come from the community-maintained fldb compilation (CC-BY), covering 1,158 championship races from 1950 through the 2026 season. Drivers completing at least 90% of the race distance receive official positions even if they stop early. Other retirements carry no position and enter our protocol tied at the back of the field. Shared drives in the 1950s produce genuine dead heats, which are treated as ties.

Evaluation is prequential. Races are processed in date order; the first 20% are unscored warm-up; each later race is predicted before it is observed. We report log loss, Brier score and accuracy for the winner prediction, Kendall’s tau between predicted and observed orders, expected calibration error (ECE) over pooled win probabilities, and a rank-PIT statistic that measures whether the predicted distribution over finish positions matches the realized positions.

3 A rating system that accepts any noise density

The vehicle for the experiment is a Thurstonian rater built on a discrete lattice of ability values. Each driver’s belief is a probability density on the lattice, unrestricted by any parametric family. A race performance is ability plus noise, with the noise drawn from a base density that is simply an array on the lattice: this is what makes the experiment of Section 4 possible. Between races, beliefs diffuse in proportion to elapsed time.

Prediction convolves each belief with the base density and computes win probabilities as exact order statistics of the field, ties included. Updating computes, for each driver, the exact likelihood of the observed finish order as a function of that driver’s ability, holding opponents at their predictive densities; a forward and a backward pass over the finishing order make this an $O(N)$ computation per race, validated against brute-force simulation.

Exactness matters at Formula 1 field sizes. Standard systems reduce a finish order to pairwise or adjacent comparisons, and a twenty-car grid implies 190 correlated pairwise results. Under the protocol of Section 2, with the noise still Gaussian:

system	log loss	accuracy	tau	ECE	rank-PIT
Thurstone lattice	2.225	.316	.328	.015	.176
TrueSkill	2.346	.275	.300	.011	.199
Elo (multi-entrant)	2.465	.303	.319	.026	.220
OpenSkill Plackett–Luce	2.705	.259	.283	.023	.223
OpenSkill Thurstone–Mosteller	2.951	.007	.106	.010	.628
uniform baseline	3.162	.043	—	—	—
Glicko-2	3.727	.271	.255	.022	.166

Glicko-2 scores below the uniform baseline: converting a grid into nineteen results within one rating period destabilizes its volatility estimation, and our implementation includes stabilizing guards without which it does not complete the dataset. OpenSkill’s all-pairs variant diverges similarly. The lattice system is therefore a credible baseline, and every improvement reported next is measured against its own Gaussian configuration, so the noise experiment is not confounded with the choice of system.

4 The noise density experiment

The motivated candidate follows directly from the physics. With probability $1 - p$ the noise is standard Gaussian, representing pace on a day the car survives. With probability p it is drawn uniformly from a wide interval far on the slow side, representing retirement. Figure 1 shows the shape.

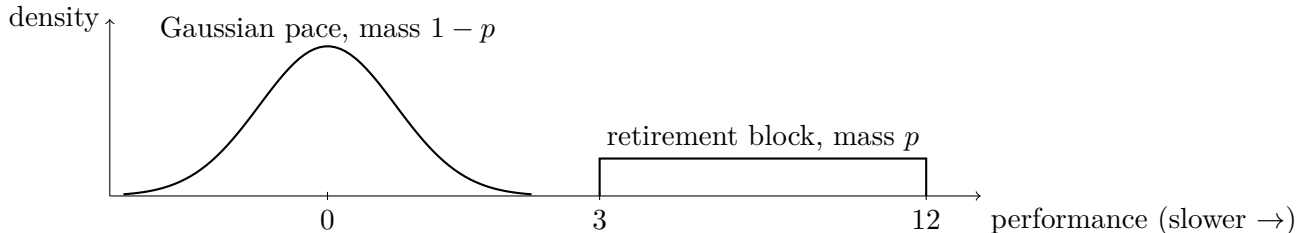


Figure 1: The two-component base density: a continuous Gaussian pace component and a separated rectangular block of slow mass representing retirement.

We compare this density against the Gaussian default and against generic departures from normality: symmetric heavy tails, skew, and a concentrated pace core. All configurations differ only in the base density.

base density	log loss	ECE	rank-PIT
era-adaptive block	2.200	.016	.185
$N(0, 1)$ plus 25% block	2.203	.015	.158
skew-normal ($a=2$)	2.222	.010	.165
Gaussian	2.225	.015	.176
tight core (0.5) plus 25% block	2.254	.013	.136
Student- t ($\nu=5$)	2.271	.018	.171
near-Dirac core plus block	2.42–2.52	—	—

The block improves every metric relative to the Gaussian. Because a fixed 25% mass is wrong for most decades of the table in Section 1, the era-adaptive variant estimates p from a trailing window of observed retirement frequency, using past races only. It attains the best log loss and rank correlation of any configuration tested, and its fitted mass declines across the dataset to finish near the modern retirement rate.

The generic departures fail, which sharpens the conclusion. Symmetric heavy tails hurt increasingly as tails fatten, because symmetry places equal mass on implausibly fast performances. Skew-normal noise changes little, consistent with two facts about rank data: the difference of two independent identically distributed noises is symmetric whatever their skew, and ranks are invariant to monotone transformations of the time scale. A near-Dirac pace core loses about 0.2 of log loss, showing that variation from strategy, traffic and weather is comparable in size to ability differences. The value is not in non-normality as such. It is in matching the actual structure of the noise, here a separated mixture component with a physical meaning, and only a system with a free noise density can express that.

5 Comparison with betting odds

As a secondary exercise we compare the ratings with bookmaker prices. No public archive of Formula 1 winner odds exists; from Wayback Machine captures of odds-comparison pages we recovered full-field consensus probabilities for 144 races between 2008 and 2025 (medians across roughly twenty bookmakers, normalized to remove the margin). Snapshot timing matters: a price taken after qualifying incorporates the grid and is not a baseline for any forecast made without it. We therefore stratify the 135 covered post-warm-up races by snapshot date and match information tiers, comparing pre-qualifying prices against results-only ratings and race-day prices against ratings that observe each grid as an additional ranking event before predicting.

stratum	races	market	tier-matched ratings	gap (se)
pre-qualifying	88	1.581	2.177	+0.60 (.08)
qualifying day	26	1.622	1.820	+0.20 (.11)
race day	21	1.310	1.625	+0.31 (.14)

The stratification shows the market’s advantage is mostly not the grid: the full gap is already present in pre-qualifying prices, which reflect current car performance, testing and practice pace. Grid information matters (observing the grid improves the ratings from 2.177 to 1.926 on the pre-qualifying stratum) but does not close the gap, and remarkably the pre-qualifying market outperforms even ratings holding the realized grid, information those prices could not contain. Pooling tests confirm the point: on the pre-qualifying stratum every rating system considered in Section 3, including the grid-aware variant, receives a fitted weight of zero alongside the market price, and the fitted market temperature does not validate out of sample at this sample size. Within the limits of the record, Formula 1 winner markets appear efficient with respect to results-based ratings at every information tier we can construct.

6 Limitations

Constructor effects are folded into driver ratings; separating driver from car is the natural next study. Retirements enter tied at the back, which discards the official ordering of retirees by laps completed. The ratings are one-dimensional, with no axis for wet weather or circuit type. The odds record is as described in Section 5, and in particular the grid position asymmetry means the market comparison understates what a qualifying-aware rating could achieve.

Reproducibility

Each table is one command from the `winning` package (github.com/microprediction/winning): `python -m winning.benchmarks.run_benchmark --dataset f1` for Section 3, and the scripts `research/f1_dirac_disaster.py`, `research/t_sweep.py`, `research/f1_era_slab.py`, `research/f1_odds_fe`, `research/f1_odds_parse.py` and `research/f1_market_test.py` for Sections 4 and 5. Race data is fetched at run time from `f1db` (CC-BY-4.0). The lattice machinery is the `thurstone` package; the underlying ability transform is described in Cotton, *SIAM J. Financial Mathematics* 12(1), 2021.